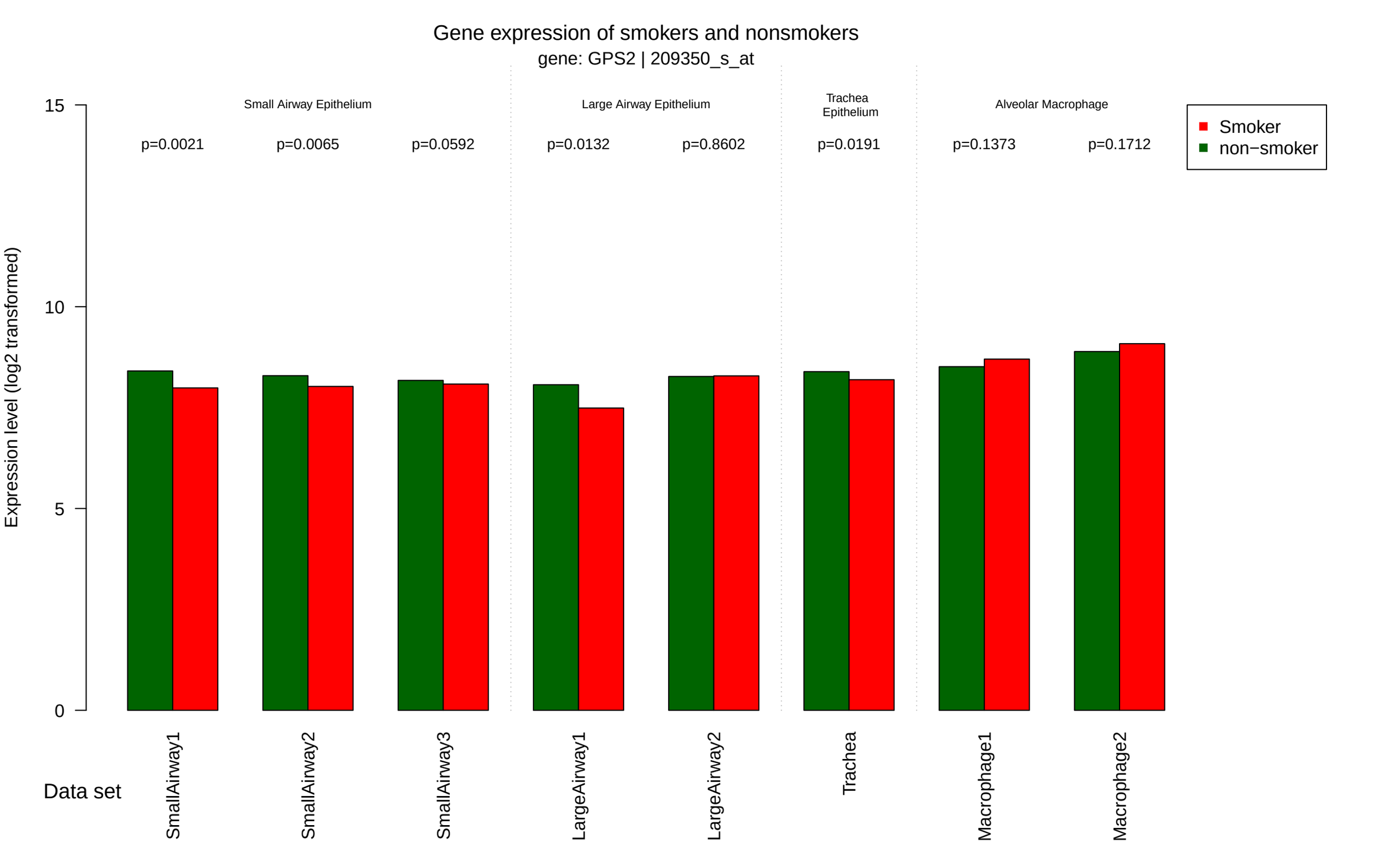
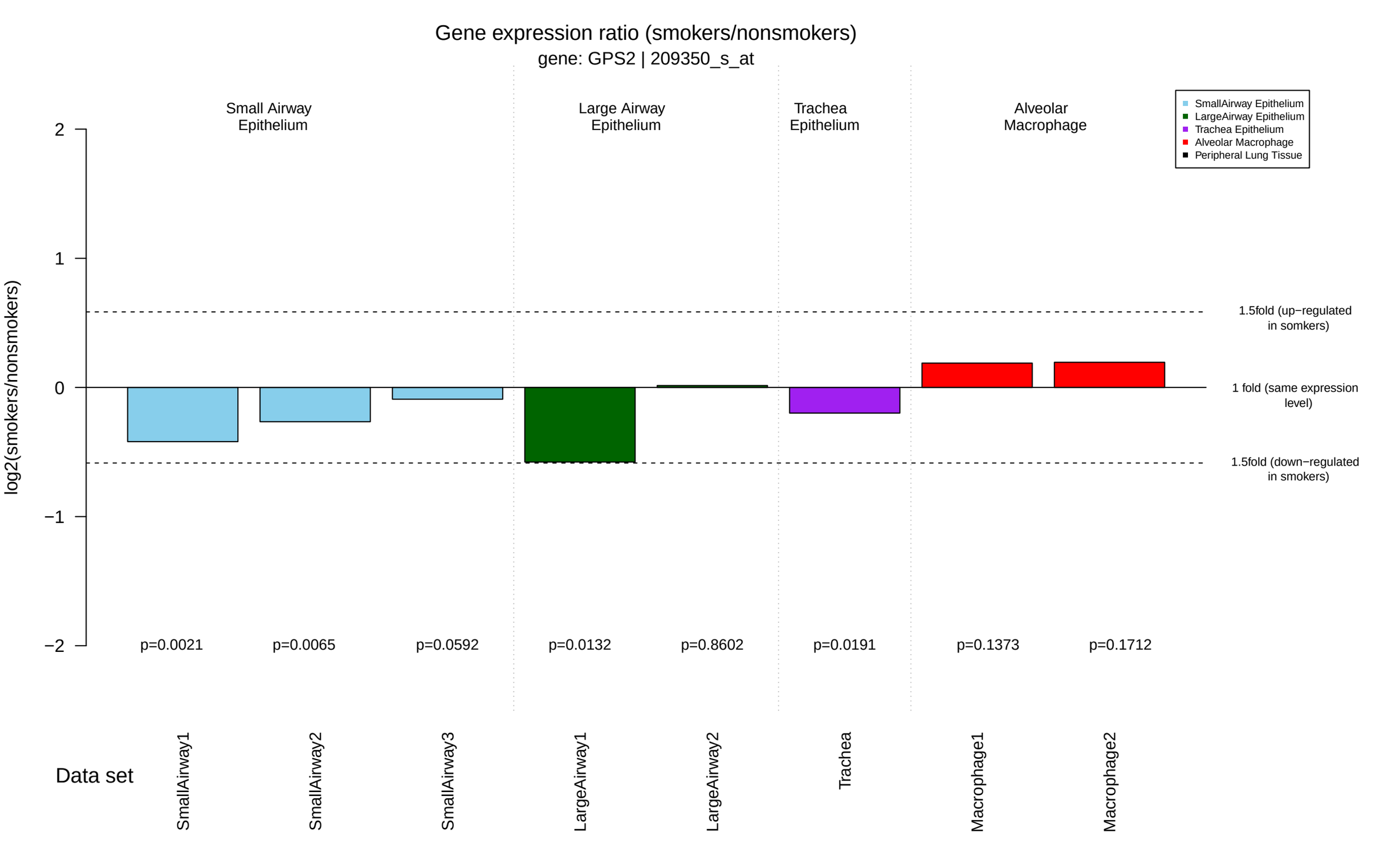
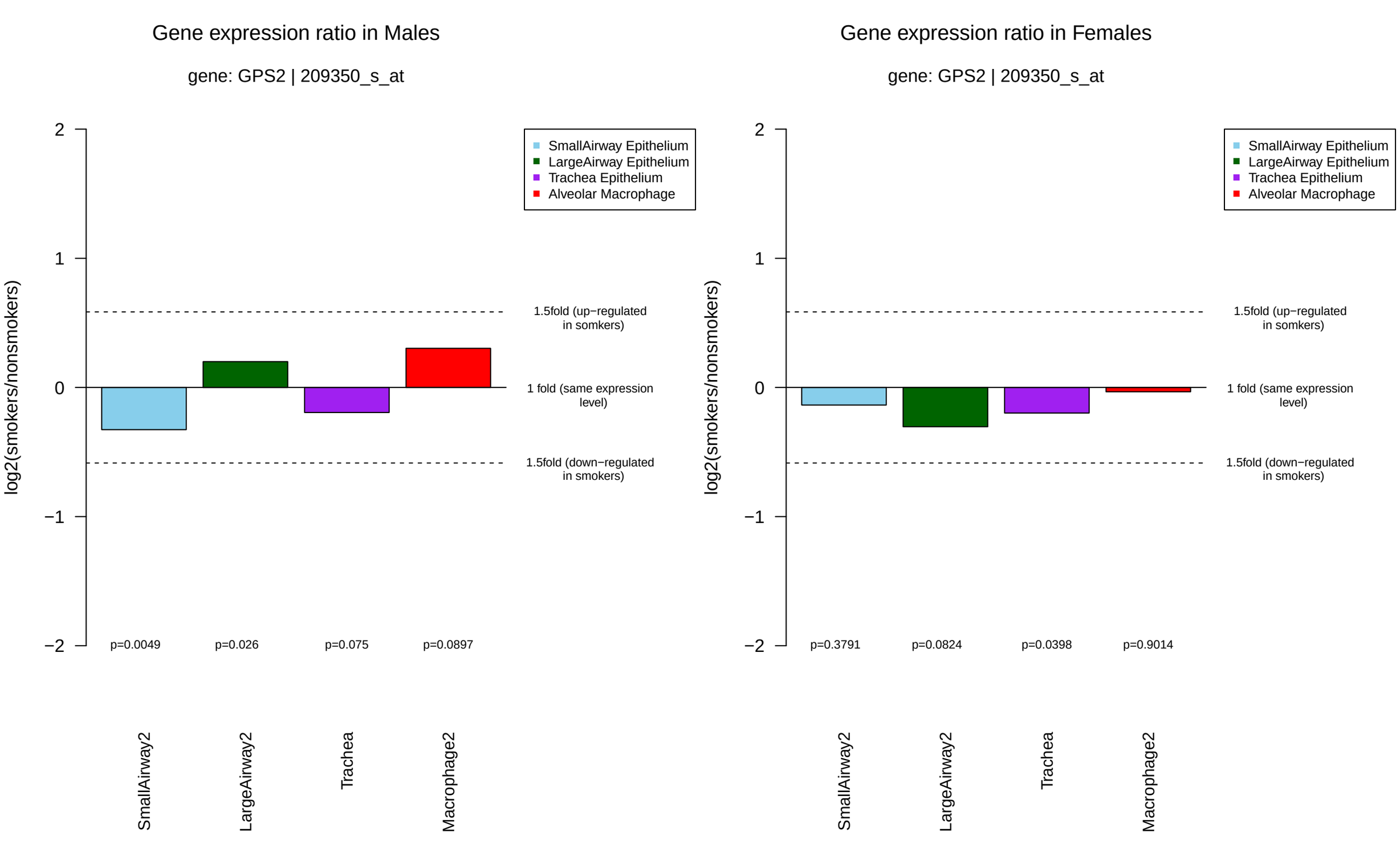
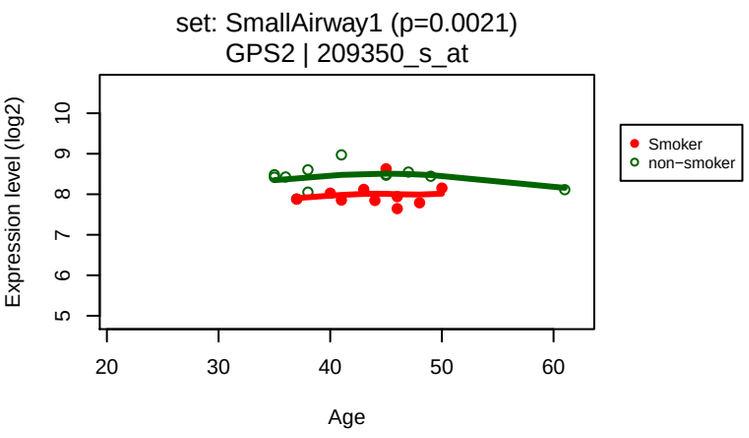
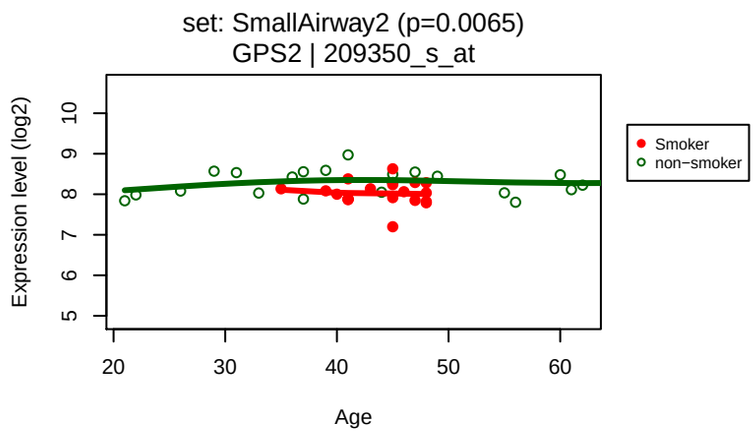
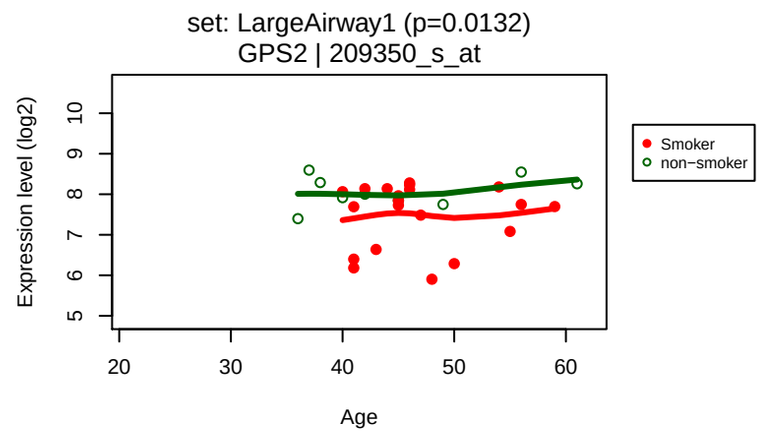

Gene expression of smokers and nonsmokers
gene: GPS2 | 209350_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0021
p=0.0065
p=0.0592
p=0.0132
p=0.8602
p=0.0191
p=0.1373
p=0.1712
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

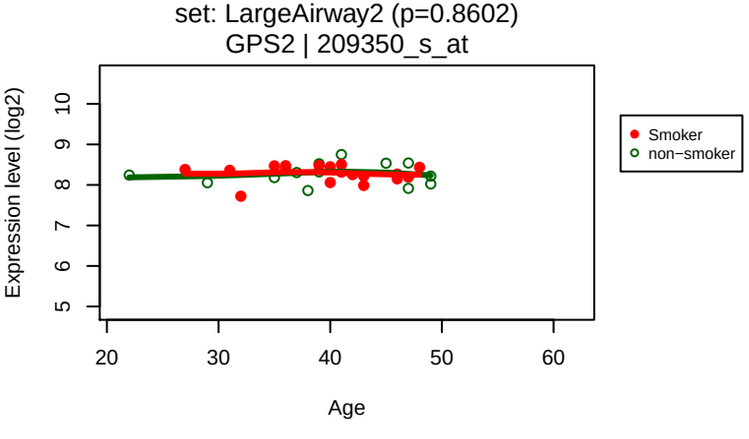

Gene expression ratio (smokers/nonsmokers)
gene: GPS2 | 209350_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0021
p=0.0065
p=0.0592
p=0.0132
p=0.8602
p=0.0191
p=0.1373
p=0.1712
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

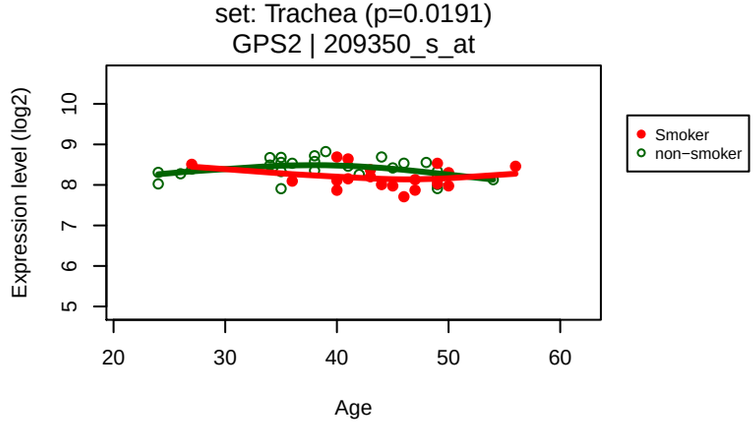

Gene expression ratio in Males
gene: GPS2 | 209350_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0049
p=0.026
p=0.075
p=0.0897
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPS2 | 209350_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3791
p=0.0824
p=0.0398
p=0.9014
SmallAirway2
LargeAirway2
Trachea
Macrophage2

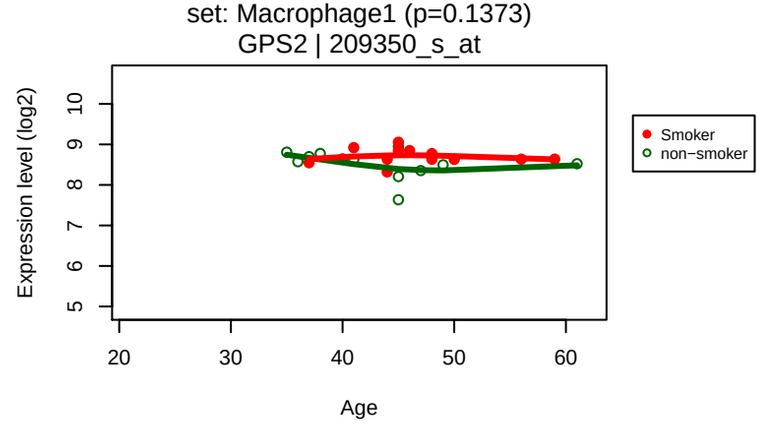

set: SmallAirway1 (p=0.0021)
GPS2 | 209350_s_at
Expression level (log2)
Age
Smoker
non−smoker

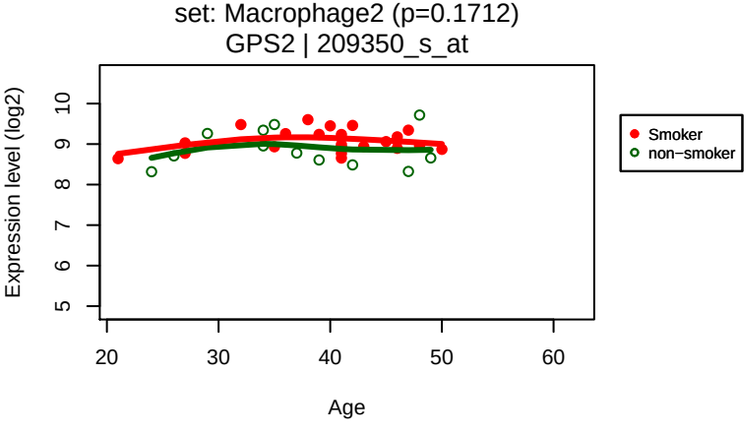

set: SmallAirway2 (p=0.0065)
GPS2 | 209350_s_at
Expression level (log2)
Age
Smoker
non−smoker

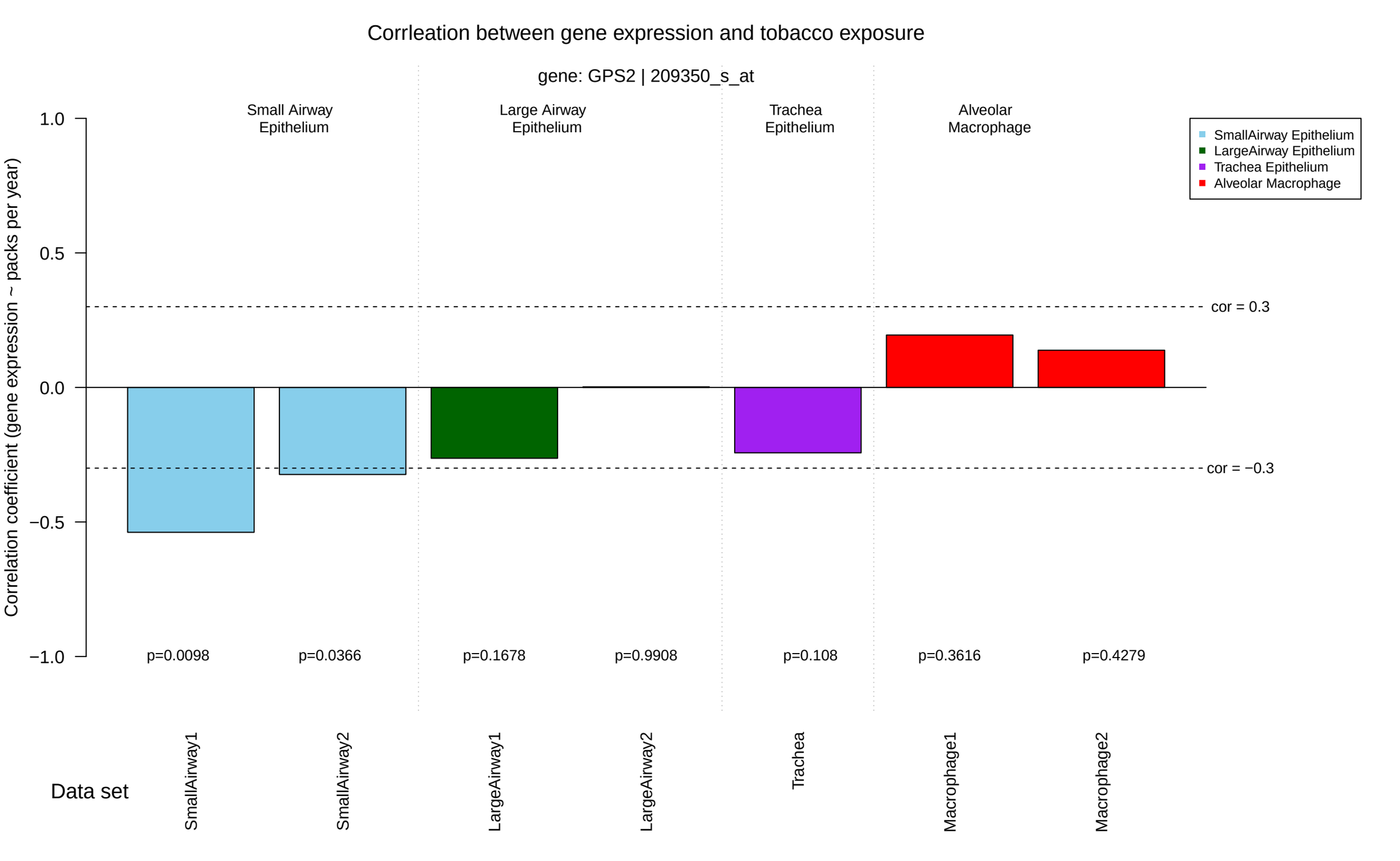

set: LargeAirway1 (p=0.0132)
GPS2 | 209350_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8602)
GPS2 | 209350_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0191)
GPS2 | 209350_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1373)
GPS2 | 209350_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1712)
GPS2 | 209350_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPS2 | 209350_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0098
p=0.0366
p=0.1678
p=0.9908
p=0.108
p=0.3616
p=0.4279
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2